

Nestor Rendon, PhD

| AI Researcher | Data Science | LLM

Lugano, Switzerland

✉ nestor.rendon@udea.edu.co • [in nestor-rendon-617103156](#) • [🌐 NestorRendon](#)

Professional Profile

AI Researcher with a PhD in Data Science and Statistical Modeling. Experienced in designing and implementing intelligent systems, focusing on modular ML pipelines, conversational agents, LLM finetuning, and scalable AI deployment in production environments. Strong background in unsupervised learning (audio processing), deep learning (Vision), and high-dimensional data modeling. Current research interests include software architectures for multi-agent systems and the integration of Large Language Models into modular, maintainable systems.

Professional Experience

XFarm Technologies

Vision engineer- Data Scientist

Switzerland

2020–Present

- Design and implementation of conversational agents and deep learning components for large-scale agricultural monitoring.
- Development of end-to-end ML pipelines encompassing data ingestion, preprocessing, training, validation, and deployment.
- Design of modular software components for integrating AI vision models into production systems.
- Monitoring of inference performance and scalability in cloud environments (AWS, GCP).
- Experimentation with multi-agent and agent-based orchestration frameworks for intelligent system coordination.
- Utilization of containerization, version control, and structured development workflows.

University of Antioquia

PhD Researcher – Machine Learning & Statistical Modeling

Colombia

2021–2024

- Development of novel unsupervised learning methodologies for soundscape analysis.
- Proposal of internal clustering validation methods based on Fréchet distance.
- Design and implementation of modular experimental frameworks for large-scale data analysis (Geospatial, Audio, and Time-Series data).
- Statistical modeling of high-dimensional datasets.
- Publication of peer-reviewed articles in international AI and applied research journals.

Laboratoire MIA

Machine Learning Engineer

France

2022–2023

- Development of benchmarking frameworks for unsupervised learning algorithms.
- Experimental evaluation and implementation of modular ML pipelines for ecoacoustic data.
- Collaboration in interdisciplinary AI research projects.

SUPSI - DTI (Reteca foundation internship)

Machine Learning Engineer

Switzerland

2020–2021

- Implementation of deep learning pipelines for object detection systems.
- Development of software modules for data acquisition and preprocessing.
- Collaboration with industrial partners on applied ML solutions.

IES and Colombia de Colores corporation

Software ML engineer and Teacher

Colombia

2017–2020

- Development of software for Gambling machines.
- Development of ML modules for rain detection in acoustic recordings.
- Programming Teacher.

Education

University of Antioquia (Cum Laude Distinction)

PhD in Electrical and Computer Engineering: Data Science & Statistical Modeling 2024

University of Antioquia

MSc in Machine Learning 2020

University of Antioquia

BSc in Electronic Engineering 2017

Selected Publications

Rendon et al. (2025). *Letting Ecosystems Speak for Themselves: An Unsupervised Methodology for Mapping Landscape Acoustic Heterogeneity*. Environmental Modelling & Software

Rendon et al. (2023). *Uncertainty Clustering Internal Validity Assessment using Fréchet Distance*. Engineering Applications of Artificial Intelligence.

Rendon et al. (2022). *Automatic Acoustic Heterogeneity Identification*. Ecological Indicators.

PhD Thesis url: <https://drive.google.com/file/d/1JF-qFjJJAmMfALrwNTy0mNbJbRv53Uj7/view>

Other publications available on my Google Scholar

Technical Skills

Programming: Python (advanced), C++, R, SQL, Bash

AI & ML: LLM agent, Deep Learning, Unsupervised Learning, Statistical Modeling, Clustering

Systems & Architecture: Modular system design, AI model integration, Multi-component systems, CI/CD

Frameworks: Agent Development Kit (ADK), PyTorch, TensorFlow, FastAPI, Scikit-learn, OpenCV, Pytest

Cloud & DevOps: AWS, Docker, ML pipelines, Linux, Git, CUDA, Grafana, Power BI, Metabase.

Research Methods: Experimental design, statistical validation, benchmarking, Documentation (Sphinx)

Languages

Spanish (Native)

Italian (C1)

English (B2)

French (A2)

References

Prof. Abel Paz

AI leader - xFarm technologies Spain

abel.paz@xfarm.ag

Prof. Thierry Bouwmans

Université de La Rochelle -France

thierry.bouwmans@univ-lr.fr

Prof. Claudia Victoria Isaza

University of Antioquia- Colombia

victoria.isaza@udea.edu.co